THE COGNITIVE BICYCLE

Honest AI Use, Intellectual Ownership, and the Framework the Academy Needs

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Abstract

Generative artificial intelligence has altered the mechanics of scholarly production without eliminating human responsibility for intellectual direction. This document argues that integrity in modern scholarship depends not on rejecting new tools, but on documenting their use transparently. The central thesis is that scholarly trust rests on accurate lineage — the traceable connection between human intention, methodological development, and final output. Drawing on empirical research into disclosure behaviour, historical precedent, and a structured evaluative framework for intellectual lineage, this work proposes lineage reporting as a constructive institutional response to AI-assisted intellectual work and demonstrates how such thinking may be applied in practice. The lineage framework referenced in this paper has previously been developed as a working manuscript and is cited here as foundational conceptual work (Keen, 2026).

Part I — The Relocation of Skill

1.1 What the Luddites Got Wrong

The Luddites were not simply afraid of machines. They were skilled craftsmen who understood that the power loom threatened the economic and social structures within which their expertise had value. Their resistance was rational within their existing framework. What they could not fully anticipate was that the framework itself would change, not the machine.

The historical error was not resistance itself. It was the assumption that the value of the work resided primarily in the physical act of weaving rather than in the knowledge of pattern, material, and judgement. When the loom replaced the manual act, it appeared that the craft had been stolen. In reality, the craft had been relocated.

Technological change repeatedly shifts where skill resides rather than eliminating skill entirely. The skill that survived mechanisation was not the movement of the shuttle but the capacity to determine what to produce, why to produce it, and how to judge the result.

1.2 The Same Structural Question in Academia

Some contemporary academic responses to generative AI resemble historical transitions in which established measures of labour were preserved even as the nature of work evolved.

As generative systems have become capable of producing fluent, structured prose, scholarly evaluation has often remained tied to the mechanics of text production. The underlying assumption, often implicit, is that authorship is primarily demonstrated through physical writing labour.

Yet if intellectual value resides in identifying problems, structuring arguments, and defending conclusions, textual production alone may not be the most reliable indicator of scholarly contribution. This resembles the loom argument and may repeat similar structural assumptions: that the value of the craft resides in the physical act rather than in the knowledge and judgement that directs it.

The central evaluative question may therefore be shifting from ‘who produced the text?’ to ‘who directed the thinking?’

1.3 What Current Policy Measures

Policies at major journals and publishers demonstrate increasing recognition of AI-assisted workflows. Journals such as Science (AAAS), Nature, and IEEE publications have adopted policies addressing AI-assisted writing, typically restricting AI authorship while permitting limited assistance under disclosure requirements.

These policies primarily regulate production mechanics rather than documenting intellectual lineage. The question that remains insufficiently addressed is not whether tools were used, but whether the intellectual lineage of the work remains transparent and defensible.

> *The evaluative framework proposed in this paper is designed to structure human judgement rather than replace it. It does not claim to mechanically determine authorship. Instead, it provides a set of defined dimensions — origin, inheritance, descent, and sanction — through which intellectual lineage and responsibility may be assessed.*

Part II — The Crisis of Disclosure

2.1 The Empirical Gap

The scale of undisclosed AI use in academic publishing has begun to receive systematic attention. A large-scale study published in the Proceedings of the National Academy of Sciences analysed more than 5.2 million academic papers across 5,114 journals and identified patterns consistent with rapidly increasing AI-assisted writing across disciplines, while disclosure rates remained extremely low (He & Bu, 2026).

In specialised domains, similarly low disclosure rates have been documented. Bibliometric analysis in radiology literature identified disclosure rates near 1.7%, despite increasing evidence of AI-assisted manuscript preparation (Barrett et al., 2025).

Taken together, these findings indicate a measurable gap between technological adoption and formal reporting behaviour.

2.2 Structural Incentives and Incomplete Lineage

Observed underreporting should not be interpreted solely as individual failure. It may reflect structural misalignment between evolving workflows and inherited disclosure models.

Binary reporting structures compress complex workflows into simplified categories. Where partial assistance is difficult to classify, silence may appear administratively safer than transparency. Such environments create incomplete lineage documentation across the scholarly record — not because researchers lack integrity, but because institutional frameworks may not yet fully accommodate contemporary workflows.

2.3 The Fabrication Principle

Engineering practice offers a clarifying parallel. Reliable systems rarely emerge from uninterrupted theoretical success. They develop through iterative correction:

`design → failure → correction → documentation`

Fabricators adjust structures until performance is achieved. Drawings are subsequently revised to reflect working reality. When early failures are omitted from documentation, the final structure appears correct from inception. The system functions. The lineage weakens. Future engineers inherit simplified histories that conceal the actual pathway to reliability.

A similar phenomenon may occur in scholarship. When developmental stages are omitted from the historical record, the resulting work appears artificially linear. This phenomenon may be described as retrospective coherence — the reconstruction of intellectual history as if outcomes were inevitable. Transparent lineage documentation preserves intellectual continuity.

2.4 Illustrative Lineage Analysis

The following demonstrates how structured lineage reasoning may be applied to observed disclosure patterns. This is presented as an illustration of the evaluative structure rather than a system-wide judgement.

ILLUSTRATIVE OBSERVATIONS — OBSERVED DISCLOSURE PATTERNS

Origin: *May become incomplete where production pathways remain undisclosed*

Inheritance: *Intermediate reasoning steps often remain undocumented*

Descent: *Continuity records may weaken when workflows are implicit*

Sanction: *Institutional standards vary widely across domains*

Lineage: *Verification becomes difficult when components remain unclear*

The purpose of this illustration is clarification, not adjudication. Where disclosure norms address compliance rather than lineage, a declaration that AI was used satisfies a procedural requirement but does not constitute a lineage record. The distinction between these two things is where the framework's practical contribution lies.

Part III — The Cognitive Bicycle

3.1 Race or Journey?

If a person must travel from one place to another and a bicycle is available, using it is not deceptive. The bicycle is more than a tool for efficient travel; it is a technology that reconfigures human possibility. It collapses distances that had defined isolation, allowing thought and relationship to travel beyond the walking radius of solitary, unaided effort.

There is a caveat: unless he was in a running race. In a race, performance depends on unaided exertion. In a journey, success depends on reaching a meaningful destination.

Scholarship functions more accurately as a journey than as a race. Its measure is not endurance of transcription. It is understanding. The defence of a doctoral thesis is an examination of the explorer's knowledge of the territory, not an audit of his boots.

3.2 The Bicycle as Cognitive Infrastructure

A large language model functions as conceptual infrastructure. It enables thinkers to traverse connections across domains that would otherwise require separate years of isolated study. It does not determine conclusions. It enables exploration.

Tool presence does not determine authorship. Direction determines authorship.

3.3 The Scholar as Rider

The rider provides what the bicycle cannot. The destination: the original insight that constitutes the scholarly contribution. The route: the sequence of arguments, applications, and examples. The steering: continuous judgement, correction, and selection of what is valid. The map: the integrative mind that holds the whole territory together and can navigate it independently.

The bicycle expands movement. The rider determines meaning.

3.4 Access and Intellectual Infrastructure

Formal scholarly systems provide structured intellectual support: libraries, supervisors, seminars, peer networks, and institutional resources. These supports are not equally accessible. Independent researchers may lack consistent access to the apparatus that shapes, challenges, and amplifies thinking at well-resourced institutions.

Used transparently, generative systems may provide partial equivalents — enabling exploration, stress-testing of reasoning structures, and cross-disciplinary connection. Transparency preserves fairness. The researcher who declares this use and can defend their work without it has not compromised their scholarship. They have documented the tools that substituted for institutional access.

Part IV — The Framework

4.1 Four Tests of Intellectual Ownership

To evaluate ownership in AI-assisted workflows, four operational tests are proposed. These are not novel inventions. They formalise questions that examiners have long applied in viva voce settings. The framework makes them explicit and portable for an environment in which traditional signals of authorship are no longer reliable.

Spark Test

Where did the initiating idea originate? The candidate must identify a moment of human inquiry that contains the generative spark — a question, observation, or pre-AI insight. This is not a demand for a manuscript predating all AI involvement. It is a demand for evidence that the destination was chosen by a human mind.

Map Test

Who structured the argument? The candidate must demonstrate that the architecture — which domains are connected, which arguments are central — was directed by them, not generated by the tool.

Voice Test

Whose judgement governs the reasoning? Every substantive intellectual contribution is shaped by a perspective — values, commitments, and limiting principles that are the author's own. These must be identifiable and defensible without AI-generated formulation.

Defense Test

Can the work be defended independently of the tool? This is the oldest test of academic ownership and the one that AI use, honestly declared and rigorously directed, does nothing to compromise. The candidate who used AI to articulate a framework they understood before the articulation can defend it without the tool. The candidate who used AI to generate a framework they did not understand cannot.

4.2 The Lineage Declaration

AI-assisted work should include structured provenance documentation — a Lineage Declaration — recording the concept origin; the scope and stages of tool involvement; the boundaries of assistance; and the author's independent defence capacity.

This functions as intellectual engineering documentation. It converts invisible workflow into visible lineage. It does not justify tool use. It records intellectual development. A dissertation arguing for transparent lineage that conceals its own lineage would fail its own first test. The declaration is the framework applied to itself.

4.3 Integrity Questions for AI-Assisted Work

Three operational questions extend the four tests and may be applied as self-assessment or examiner prompts:

- Is the core idea genuine — traceable to identifiable human inquiry?
- Is tool use transparent — declared with sufficient specificity to constitute a lineage record?
- Can the work be defended independently — without reliance on AI-generated explanation?

4.4 Evaluation Practice

Structured challenge — rather than automated detection — may provide stronger evaluation methods as AI-generated prose becomes indistinguishable from human-authored text. Detection tools have documented reliability problems at scale. The four tests, applied through viva examination or structured review, assess intellectual ownership directly rather than inferring it from stylistic signals.

Final judgement remains human. AI may assist examiners in generating challenges, identifying structural inconsistencies, or broadening the question space. It must not determine outcomes.

4.5 Institutional Responsibility

Institutions play a stabilising role in knowledge systems. Their responsibility is not to eliminate innovation risk, but to document adaptation accurately. Responsible adoption requires acknowledging transitional phases, recording enabling methods, and preserving developmental lineage.

When institutions move from resistance to adoption to normalisation without documenting the transition, they enact precisely the retrospective coherence that erodes scholarly trust. Reliable knowledge emerges from recorded correction, not from uninterrupted apparent accuracy. Trust depends on visible developmental history.

Part V — Counterarguments and Limits

Analytical rigour requires engagement with the strongest objections. Three boundary conditions require acknowledgement.

AI Novelty

Generative systems may occasionally produce conceptual arrangements without explicit human anticipation. Where this occurs, attribution of intellectual ownership becomes genuinely ambiguous. The framework's response is that unexpected outputs do not become intellectual property by virtue of appearing. The human who recognises value in an unexpected output, integrates it coherently, and can defend its relevance is exercising the judgement that constitutes scholarship. The Defense Test will surface cases where this is not true.

Writing Cognition

The physical act of writing may support reflective cognition. Reducing transcription effort may alter thinking patterns in ways that are not yet fully understood. This is a legitimate empirical question. The framework does not dismiss it. The concern applies differentially: a candidate who uses AI to articulate an argument they have already worked through has not bypassed the cognitive labour in question. The Defense Test distinguishes these cases.

Access Inequality

Advanced AI tools may introduce new disparities rather than resolve existing ones. Institutions with greater computational resources or AI literacy may gain advantage. This objection is acknowledged and has force. The framework's claim is not that AI has equalised access, but that declared, honest AI use by an independent researcher is not categorically less legitimate than the undeclared institutional advantages available to well-resourced scholars. The inequality objection strengthens the case for transparent disclosure norms; it does not invalidate the framework.

Each of these conditions remains subject to empirical investigation. They define the framework's current boundaries rather than invalidating its structure.

Part VI — Illustrative Application to This Work

The framework is applied here to its own production. This is not a claim of verified status. It is a demonstration of what structured lineage reasoning looks like when applied to a known case — the author's honest account, offered for examination.

ILLUSTRATIVE LINEAGE ANALYSIS — THE PRESENT WORK

Spark: *Human-origin conceptual inquiry. The generative questions — file-as-species, lineage over fixity — originated in the author's prior work on digital preservation and are documented in conceptionproof.pdf.*

Map: *Human-directed architecture. The structure of this work was designed by the author. AI elaborated under direction; it did not determine the arrangement of ideas.*

Voice: *Human ethical position. The stance — that honest documentation is more valuable than the performance of unaided labour — derives from the author's lived experience.*

Defense: *Independent explanation capacity. The author can explain, defend, and critique every element of the framework without reliance on AI.*

Lineage: *Documented and acknowledged. Transformation recorded. Drift within declared bounds.*

A large language model was used in the production of this work. Its role was: research assistant, providing access to literature otherwise inaccessible to an independent researcher; critical interlocutor, stress-testing arguments under authorial direction; drafting instrument, articulating concise prompts into structured prose; and disciplinary bridge, connecting archival science, philosophy, and institutional theory.

It did not originate the ideas. It did not determine the structure. It cannot defend the argument. It was a cognitive bicycle, ridden to a destination the author chose, through territory the author can navigate without it.

Conclusion

Generative tools alter scholarly mechanics but not intellectual responsibility. The challenge is not technological control. It is lineage preservation.

The academy faces a decision point that resembles the one faced by the textile industry in the early nineteenth century. A tool has arrived that performs the physical act previously used as the primary measure of the underlying skill. The question is whether to defend the measure or clarify the skill. The loom inherited the shuttle. It could not inherit the understanding of pattern, material, and purpose. Generative AI has inherited the production of prose. It cannot inherit the identification of a problem, the construction of a framework, or the accountability for the result.

Using a cognitive bicycle to reach a destination chosen by the researcher extends discovery without diminishing authorship. Transparent documentation preserves scholarly continuity.

> *Tools expand capability. Lineage preserves truth. Where capability expands without documentation, confusion follows. Where capability expands with transparent lineage, scholarship remains reliable across generations.*

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References

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